

Sarthak Kala

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EDUCATION

University of California, San Diego

Sep 2024 – Present (Expected Dec 2025)

Master of Science, **Data Science** (GPA: 3.9/4.0)

La Jolla, CA

- **Coursework:** Safety in Generative AI, Deep Generative Models, Machine Learning, Recommender Systems, ML Systems, Advanced Data Mining

EXPERIENCE

Hewlett Packard Enterprise

Sunnyvale, CA

Machine Learning Engineering Intern

Jun 2025 – Sep 2025

- Built **self-healing network diagnostics** using **transformer-based anomaly detectors** over routing & telemetry logs; auto-triggered remediation workflows and improved fault-resolution latency by **2×**.
- Developed **multi-agent ML system** that parsed switch documentation, generated attack/validation hypotheses, and auto-translated them into executable network test code.

American Express

Gurugram, India

Senior Applied Scientist

Feb 2022 – Aug 2024

- Shipped an internal **LLM-based Marketing Q&A Copilot** using **RAG + LoRA** over campaign playbooks, brand guidelines, and measurement docs.
- Designed and deployed **Hierarchical Bayesian Marketing Mix Models** with **adstock/saturation** to measure **incremental lift** across Search, Display, YouTube, and Paid Social; ran **ROI simulations** and budget reallocation improving efficiency by **8–12%**.
- Built a unified **multimodal creative-quality + similarity system** using **CLIP embeddings** over CMS assets and platform attention signals (**scroll-stop**, **VTR**, **CTR**); trained **XGBoost** for pre-launch performance prediction and integrated **FAISS retrieval** to detect duplicate variants. Improved strong-creative recall by **15–16%** and reduced redundant uploads by **35–40%**.
- Built a **complaint segmentation system** using **Sentence-BERT** and **unsupervised clustering** to surface emerging customer pain-point themes, enabling **faster escalation** and reducing manual review workload by **30–40%**.
- Designed and implemented containerized solutions using **Docker**, orchestrated microservice architectures with **Kubernetes**, and integrated **CI/CD pipelines** to streamline deployments, ensuring enhanced scalability, maintainability, and operational efficiency.

American Express

Gurugram, India

Applied Scientist

Jun 2019 – Jan 2022

- Built **CTR/CVR models** for fixed-position MyCA referral widgets using **clickstream features** (scroll depth, dwell time, paths), device/session context, and engagement signals; increased widget CTR by **8–10%** and referral applications by **6–8%**.
- Reconstructed the full **referral funnel** and trained **Transformer models** to detect stalled referee states + approval likelihood, enabling adaptive nudges that improved approval conversion by **10–15%**.
- Developed **uplift models (X-Learner)** to target referral bonuses toward incrementally persuadable referrers; designed and ran **A/B experiment** across referral surfaces to validate incremental lift, increasing high-quality referrals by **18%** without raising promo spend.
- Built a **customer social-influence graph** via referral links + co-spend proximity; learned **Node2Vec/GraphSAGE embeddings** to identify influential referrers, increasing new-referrer participation by **~20%**.

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL, Shell, R, MATLAB, SAS, Scala

ML Frameworks: PyTorch, TensorFlow, JAX, Scikit-Learn, Hugging Face, PyMC

Distributed Systems: Spark, Kafka, Hadoop, Airflow, Kubernetes, Docker, AWS/GCP

Specialties: Deep Learning, LLMs, Recommenders, Causal Inference, Experimentation, Bayesian Modeling, Computer Vision, Embeddings, Quantization, RAG Systems